SP-14 Green Chess AI

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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Brett Talley 706-483-9553	Kenneth Burke 925-822-4937	Marc-Henry Moise 786-617-2013	Vladimir Ryzhov 470-775-1900

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1.0 Introduction

1.1 Overview

Create a self-playing AI Chess Engine that offers both player vs player and player vs

AI game modes. You may use any technology.

Objective:

Create a working chess game which follows the standard rules of chess. The chess has to have two game modes: Player vs Player (PvP) and Player vs Artificial Intelligence (PvAI). In case of abundance of time, we might decide to pursue the objective of adjustable difficulty for the AI game mode.

Code in Python using Pygame Module.

1.2 Project Scope

Working chess, meaning:

- No illegal moves
- En passant, castling

PvP chess

- White cannot move black's pieces and vice versa
- Possibly board will flip every move

PvAI chess

- Engine automatically plays a move when player makes a move

1.4 Assumptions

This will not be an online game; we will be developing the PvP section under the assumption that both players are using the same computer.

1.5 References

https://www.youtube.com/watch?v=OpL0Gcfn4B4

- Code game design for board and pieces

https://www.freecodecamp.org/news/simple-chess-ai-step-by-step-1d55a9266977/

- Creation of a simple chess ai

https://chessboardjs.com/

- Easy embedment of a chess board onto a website

2.0 Functional Requirements

2.1 General Requirements

- No illegal moves, no phasing through pieces, etc.
- En passant, castling

2.2 PvP Requirements

- 2 players
- Can only move their respective pieces

2.3 PvAI Requirements

- Engine automatically generates the best move
- If time allows it, rank the moves
- Ranking the moves allows for adjustable difficulty

3.0 External Interface Requirements

3.1 User Interface Requirements

Mandatory:

- Menu with choice of PvP and PvAI
- Chess pieces graphics

Optional (complete if there is enough time):

- Show legal moves on piece clicked
- Choice of difficulty vs AI
- Option to resign/draw